

Astrodynamics I: Final Exam

June 9, 2014 1:00PM-2:40PM (100 minutes)

Use the following constants and formulae without proof (if necessary).

- $\vec{A} \times (\vec{B} \times \vec{C}) \equiv (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$
- $\vec{A} \cdot (\vec{B} \times \vec{C}) \equiv \vec{B} \cdot (\vec{C} \times \vec{A}) \equiv \vec{C} \cdot (\vec{A} \times \vec{B})$
- $\cos v = \frac{\cos E - e}{1 - e \cos E}, \quad \sin v = \frac{\sin E \sqrt{1 - e^2}}{1 - e \cos E}$

$$\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r}$$

$$(\ddot{\vec{r}} + \frac{\mu}{r^3} \vec{r}) \cdot \vec{h} = 0$$

1. (30) Derivation of Kepler's 1st Law

- a. (15) Start from the two body equations of motion to derive the trajectory equation in vector form:

$$\dot{\vec{r}} \times \vec{h} = \mu \frac{\vec{r}}{r} + \vec{B}$$

$$\dot{\vec{r}} \times \vec{h} - \mu \frac{\vec{r}}{r} = 0$$

where $\vec{B}$ is an integration constant defined as the eccentricity vector.

- b. (15) Start from the trajectory equation in vector form to derive its scalar form:

$$r = \frac{p}{1 + e \cos v}$$

$$r = \frac{\frac{h^2}{\mu}}{1 + \frac{B}{\mu} \cos v}$$

$$r = \frac{h^2}{\mu + B \cos v}$$

where the semilatus rectum p and the eccentricity e are defined as $p = h^2 / \mu$ and $e = B / \mu$, respectively.

$$\begin{aligned} & \dot{\vec{r}} \times (\vec{r} \times \dot{\vec{r}}) \\ &= (\dot{\vec{r}} \cdot \dot{\vec{r}})\vec{r} - (\dot{\vec{r}} \cdot \vec{r})\dot{\vec{r}} \\ &= \frac{v^2 + \dot{r}^2}{r} \vec{r} - \dot{r} \dot{\vec{r}} = \mu \frac{\vec{r}}{r} + \vec{B} \end{aligned}$$

2. (25) Derivation of Kepler's Equation

- a. (10) Start from a relation between true anomaly and eccentric anomaly

$$\cos v = \frac{\cos E - e}{1 - e \cos E}$$

to derive

$$dv = \frac{a \sqrt{1 - e^2}}{r} dE$$

$$h = r^2 \dot{v}$$

$$2rv\dot{v} + v^2$$

$$h = r^2 \frac{dv}{dt}$$

- b. (15) Then, start from the constancy of specific angular momentum

$$h = r^2 \dot{v} \equiv \text{constant}$$



to derive Kepler's equation in basic form:

$$M \equiv n(t - T) = E - e \sin E$$

where $n \equiv \sqrt{\mu / a^3}$ is the mean motion and T is the time of periapsis passage.

3. (25) Observation Transformations

- a. (10) Determine the rotation matrix for transformation from ECI frame (IJK) to Topocentric frame (SEZ). Determine all the elements of the matrix explicitly.
- b. (15) For an Earth-orbiting satellite, a pair of position and velocity vectors $(\vec{r}_{IJK}, \vec{v}_{IJK})$ in the ECI coordinate system and the site location is known by its location parameters $(\vec{r}_{SiteIJK}, \phi_{gd}, \theta_{LST})$. Describe the procedure of determining range (ρ), azimuth (β), elevation (e), and their time rates.

4. (20) Discuss the following terminologies succinctly and precisely:

- a. (4) Geodetic Latitude & Astronomical Latitude
- b. (4) RSW Coordinate System (Satellite-Based Coordinate System)
- c. (4) NTW Coordinate System (Satellite-Based Coordinate System)
- d. (4) Mean Solar Time
- e. (4) Mean Sidereal Time